

Assignment

Case Study: LFP Cell Physical Modeling & Parametrization in PyBaMM

Background:

PyBaMM (Python Battery Mathematical Modelling) is an open-source battery simulation package written in Python. In PyBaMM, it is possible to simulate high fidelity cell models such as Single Particle Model (SPM) or Doyle-Fuller Newman (DFN) model. There are many different available parametrizations for different chemistries that can be used to simulate a specific cell behavior. In this case study, you are expected to build an DFN model in PyBaMM by using the built in functionalities of PyBaMM. The cell to be modeled is A123 ANR26650M1B (rebranded as Lithium Werks) with 2.5 Ah capacity.

In the folder, you have been provided:

- Charge data for 1C, 2C, 3C and 4C
- Discharge data for 1C
- HPPC data

The data does not cover a temperature range, so the temperature effects can be ignored.

Your tasks are:

1. Building the DFN model and selecting a PyBaMM parameter set

- Build the DFN model
- Select the most suitable parameter set from [Parameter Sets](#) or you might find an open source parameter set.

2. Simulate the model

- Apply the current profile from the HPPC data and simulate the model
- Plot and compare the simulated voltage response by the measured voltage from the HPPC data provided.
- Calculate error metrics between HPPC voltage and simulated voltage such as root mean square error and explain why you choose the error metric.

3. Analysis

- Analyze the voltage residual plots and identify when and why voltage residuals show up.

- Decide about which parameters to be optimized. These parameters can be grouped based on parameter type or priority if needed.
- If required, do manual parameter tuning on the current parameter including capacity and Open Circuit Potential(OCP) related parameters.(OCP curves from the literature can be used)
- Optionally, you can find some other external resources/datasets to improve the parametrization. If the resource you find doesn't directly belong to the A123 ANR26650M1B, state your assumptions and justify the usage.

4. Optimization

- Assess which methods can be used (gradient-based or non-gradient based) for the parameter optimization.
- Construct an optimization pipeline in Python(any optimization toolbox can be used) to decrease the error between simulated and measured voltage.
- The parameter optimization pipeline should take the charge data for 1C, 2C, 3C and 4C and discharge data for 1C as input.
- Run the optimization pipeline and get the improved set of parameters. Save the parameters.

5. Validation

- Validate the resulting parameter set for A123 ANR26650M1B cell with respect to the HPPC data provided or any other data used during the training/optimization process.
- By using the optimized set of parameters, simulate the model again and plot the important physical variables such as anode potential voltages vs Li+.
- Calculate error metrics between the simulated voltage and validation voltage such as root mean square error and explain why you choose the error metric.

Deliverables:

Code	- DFN model simulation - DFN parameter optimization - Optimized set of parameters(saved in a desired format) - DFN model validation
Plots	- Measured vs simulated voltage in step 2 - Voltage estimation error in step 2 - Measured vs simulated voltage in step 5(w.r.t. training (HPPC) data) - Voltage estimation error in step 5 (w.r.t. training(HPPC) data) - Important physical variables such as anode potential voltage.

README.md	- How to run each script / model
Report	- Document the main steps and intermediate steps, methodology and assumptions - Justify each decision taken - Summarize results & key observations - Explain and comment on the plots - Discussion
GitHub Repo	- Push all source files, data references, and documentation. - Provide the repository link.

Good luck.

Due date: 7 days (max.) from date of receipt